

## NEWPORT, AR, USA

WMO: 720377

Lat: 35.633N

Lon: 91.167W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 13917

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |            | 1%          |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-8.3 | (c)<br>-6.3 | (d)<br>-15.1                  | (e)<br>1.0 | (f)<br>-6.1 | (g)<br>-12.5 | (h)<br>1.3 | (i)<br>-4.1 | (j)<br>12.7           | (k)<br>8.2 | (l)<br>11.7 | (m)<br>8.9 | (n)<br>4.4            | (o)<br>350 | (p)<br>0.491 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 36.8            | 25.8 | 35.2 | 25.6 | 33.8 | 25.1 | 27.6                | 33.0 | 26.9 | 32.5 | 26.3 | 32.0 | 4.5                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.3                            | 21.7 | 30.0 | 25.7 | 20.9 | 29.6 | 24.8 | 19.9 | 29.0 | 87.3          | 33.0 | 84.2 | 32.6 | 81.6 | 31.7 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 11.5          | (b) 10.0 | (c) 8.7 | (e) -11.5                  | (f) 38.3 | (g) 2.5            | (h) 1.7 | (i) -13.3                                          | (j) 39.6 | (k) -14.8  | (l) 40.6 | (m) -16.2  | (n) 41.5 | (o) -18.0  | (p) 42.8 |    |    |
| (a) 11.5          | (b) 10.0 | (c) 8.7 | (e) -12.4                  | (f) 28.7 | (g) 2.3            | (h) 0.6 | (i) -14.0                                          | (j) 29.1 | (k) -15.4  | (l) 29.5 | (m) -16.6  | (n) 29.8 | (o) -18.3  | (p) 30.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.1   | 3.3   | 5.4   | 10.9  | 16.1  | 21.5  | 26.8  | 27.5  | 26.6  | 22.9  | 16.5  | 9.8   | 5.6  |
|                                                                                  | DBStd       | 9.73   | 5.72  | 6.14  | 5.88  | 4.70  | 4.10  | 2.72  | 2.56  | 2.75  | 3.42  | 4.85  | 4.98  | 5.06 |
|                                                                                  | HDD10.0     | 666    | 219   | 155   | 60    | 8     | 1     | 0     | 0     | 0     | 0     | 5     | 63    | 154  |
|                                                                                  | HDD18.3     | 1934   | 467   | 363   | 237   | 96    | 20    | 0     | 0     | 0     | 4     | 94    | 259   | 394  |
|                                                                                  | CDD10.0     | 2906   | 11    | 26    | 89    | 191   | 358   | 505   | 541   | 515   | 387   | 208   | 57    | 19   |
|                                                                                  | CDD18.3     | 1132   | 0     | 1     | 7     | 28    | 119   | 255   | 283   | 257   | 142   | 39    | 2     | 0    |
|                                                                                  | CDH23.3     | 11586  | 0     | 4     | 51    | 214   | 1012  | 2867  | 3118  | 2673  | 1255  | 376   | 17    | 0    |
|                                                                                  | CDH26.7     | 5252   | 0     | 0     | 4     | 37    | 360   | 1422  | 1492  | 1283  | 531   | 120   | 2     | 0    |
| Wind                                                                             | WSAvg       | 3.9    | 4.5   | 4.7   | 4.9   | 4.9   | 4.1   | 3.6   | 3.1   | 2.8   | 2.9   | 3.5   | 3.9   | 4.1  |
| Precipitation                                                                    | PrecAvg     | 1239   | 112   | 94    | 119   | 114   | 119   | 91    | 105   | 69    | 84    | 102   | 109   | 120  |
|                                                                                  | PrecMax     | 1873   | 208   | 198   | 252   | 291   | 265   | 155   | 208   | 111   | 206   | 310   | 232   | 259  |
|                                                                                  | PrecMin     | 842    | 44    | 29    | 34    | 11    | 8     | 27    | 27    | 7     | 0     | 4     | 22    | 8    |
|                                                                                  | PrecStd     | 228    | 44    | 45    | 53    | 58    | 62    | 38    | 51    | 32    | 48    | 71    | 62    | 62   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 19.9  | 23.3  | 27.3  | 29.2  | 33.7  | 37.4  | 37.7  | 38.7  | 34.9  | 32.4  | 26.0  | 21.3 |
|                                                                                  |             | MCWB   | 17.2  | 17.9  | 18.7  | 20.7  | 23.0  | 25.3  | 25.9  | 26.7  | 24.6  | 22.5  | 18.8  | 18.6 |
|                                                                                  | 2%          | DB     | 17.3  | 20.6  | 24.7  | 27.7  | 32.2  | 36.2  | 36.3  | 37.0  | 33.5  | 30.0  | 22.6  | 18.7 |
|                                                                                  |             | MCWB   | 14.9  | 16.6  | 18.5  | 19.6  | 22.7  | 25.1  | 26.0  | 26.2  | 24.6  | 21.5  | 17.1  | 16.5 |
|                                                                                  | 5%          | DB     | 14.6  | 17.8  | 22.4  | 26.1  | 30.4  | 34.5  | 34.8  | 34.6  | 32.2  | 27.6  | 20.2  | 16.2 |
|                                                                                  |             | MCWB   | 11.5  | 14.6  | 16.8  | 18.8  | 22.1  | 24.6  | 26.0  | 25.8  | 24.2  | 19.9  | 16.3  | 14.0 |
|                                                                                  | 10%         | DB     | 12.4  | 15.3  | 19.8  | 23.9  | 28.7  | 33.0  | 33.3  | 32.8  | 30.0  | 25.1  | 17.8  | 13.6 |
|                                                                                  |             | MCWB   | 9.6   | 12.7  | 14.9  | 17.9  | 21.7  | 24.2  | 25.9  | 25.6  | 23.1  | 18.9  | 14.3  | 11.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 17.7  | 19.2  | 20.6  | 22.8  | 25.1  | 27.4  | 28.4  | 28.6  | 26.6  | 24.5  | 20.7  | 19.3 |
|                                                                                  |             | MCDB   | 19.0  | 21.9  | 25.5  | 26.9  | 31.0  | 34.1  | 33.5  | 34.7  | 32.1  | 29.7  | 23.7  | 20.5 |
|                                                                                  | 2%          | WB     | 15.7  | 17.5  | 19.2  | 21.2  | 24.2  | 26.6  | 27.6  | 27.7  | 25.6  | 23.1  | 18.9  | 17.1 |
|                                                                                  |             | MCDB   | 17.1  | 19.5  | 23.3  | 25.7  | 28.9  | 33.3  | 32.8  | 33.2  | 31.5  | 27.9  | 21.3  | 18.2 |
|                                                                                  | 5%          | WB     | 12.6  | 15.4  | 17.9  | 20.2  | 23.5  | 25.8  | 27.0  | 27.0  | 24.9  | 21.8  | 17.0  | 14.5 |
|                                                                                  |             | MCDB   | 13.8  | 17.6  | 21.0  | 24.1  | 28.1  | 32.2  | 32.3  | 32.4  | 30.3  | 25.4  | 18.8  | 15.9 |
|                                                                                  | 10%         | WB     | 9.9   | 12.9  | 16.5  | 19.2  | 22.7  | 25.3  | 26.5  | 26.2  | 24.1  | 20.3  | 15.0  | 12.1 |
|                                                                                  |             | MCDB   | 12.1  | 14.6  | 19.3  | 22.6  | 27.1  | 31.4  | 32.0  | 31.4  | 28.5  | 23.3  | 17.6  | 13.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.3   | 9.6   | 10.6  | 11.1  | 10.5  | 11.0  | 10.3  | 10.8  | 11.7  | 12.6  | 11.0  | 8.6  |
|                                                                                  |             | MCDBR  | 12.7  | 13.3  | 13.1  | 13.0  | 12.3  | 12.5  | 12.0  | 13.1  | 13.5  | 15.1  | 13.8  | 11.6 |
|                                                                                  |             | MCWBR  | 9.8   | 9.8   | 7.8   | 6.5   | 4.9   | 3.9   | 3.8   | 4.6   | 5.5   | 7.2   | 8.7   | 9.2  |
|                                                                                  | 5% WB       | MCDBR  | 10.8  | 11.4  | 11.7  | 10.8  | 10.3  | 11.1  | 10.6  | 11.3  | 11.8  | 12.5  | 11.1  | 10.3 |
|                                                                                  |             | MCWBR  | 9.8   | 9.4   | 7.6   | 6.6   | 4.9   | 4.1   | 4.2   | 5.0   | 5.2   | 7.1   | 8.7   | 9.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.324  | 0.330 | 0.366 | 0.408 | 0.446 | 0.473 | 0.498 | 0.487 | 0.423 | 0.362 | 0.333 | 0.317 |      |
|                                                                                  | taud        | 2.464  | 2.448 | 2.378 | 2.254 | 2.214 | 2.152 | 2.121 | 2.130 | 2.300 | 2.452 | 2.486 | 2.500 |      |
|                                                                                  | Ebn at Noon | 874    | 912   | 903   | 876   | 845   | 819   | 794   | 796   | 832   | 863   | 856   | 860   |      |
|                                                                                  | Edh at Noon | 86     | 97    | 114   | 135   | 143   | 152   | 156   | 152   | 122   | 96    | 83    | 78    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.42   | 2.97  | 4.10  | 5.20  | 5.76  | 6.56  | 6.26  | 5.76  | 4.88  | 3.75  | 2.64  | 2.08  |      |
|                                                                                  | RadStd      | 0.20   | 0.39  | 0.33  | 0.37  | 0.42  | 0.42  | 0.30  | 0.37  | 0.37  | 0.46  | 0.28  | 0.23  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page